Quantitative text analysis – Essex Summer School

New forms of (model-produced) data

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Today's class

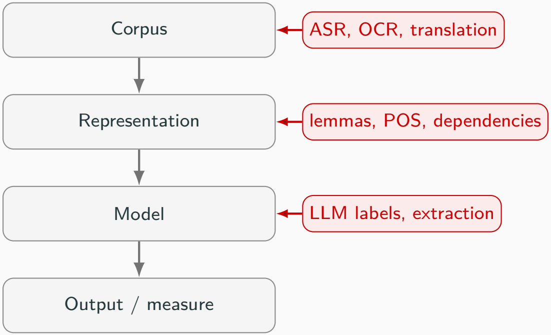
- New forms of data in quantitative text analysis
  - Transformation: machine translation and speech-to-text
  - Annotation: linguistic structure such as lemmas, POS tags, and dependencies
  - Interpretation: LLM coding, extraction, and structured labels
- 2 Flash talks: Ipek and Yizhou
- Lab session: NLP processing with House of Commons speeches

The growing importance of model-based text in QTA

Text analysis projects increasingly rely on data that first passes through a system or model

But how can we validate these model-generated outputs for our research?

- Validation depends on (1) where the model enters the pipeline, and (2) what the model does



Three ways models make data

Type	What the model does	Examples	Role in the pipeline
Transformation	changes the form of the source	translation, ASR, OCR	makes source material usable as text
Annotation	adds linguistic structure	lemmas, POS tags, dependencies	enriches text before analysis
Interpretation	applies a substantive code	LLM labels, summaries, extractions	produces a coded output for analysis

In this lecture, transformation and annotation mainly help construct the input we analyze. Interpretation is closer to measurement: the model is asked to decide what the text means for a substantive concept.

Transformation: multilingual text analysis

Comparative research often asks whether the same concept appears across languages and political contexts.

- Parties discussing migration in German, Dutch, English, and French
- News coverage of climate change across countries
- Parliamentary debates in multilingual institutions

Problem: words, grammar, idioms, and political associations do not line up neatly across languages.

Why multilingual text is hard

- Languages differ in script, morphology, syntax, and word order
- The same political concept may be expressed with different kinds of words or phrases
- Translations may preserve literal meaning but not contextual meaning
- Some expressions have no clean equivalent

Licht and Lind (2023): the goal is to obtain comparable measurements for documents that are similar at the conceptual level, even when they are written in different languages.

Running example for multilingual analysis: housing insecurity

Suppose we want to compare how parties discuss housing insecurity in English, Dutch, and German parliamentary debates.

Language	Possible terms or phrases
English	rent, evictions, homelessness, housing costs
Dutch	huur, huisuitzetting, dakloosheid, woonlasten
German	Miete, Zwangsraeumung, Wohnungslosigkeit, Wohnkosten

- we do not just want matching words but comparable measurements of the same concept across languages.

Licht and Lind (2023): three strategies

Strategy	Basic move	What gets compared?
Separate analysis	analyze each language separately	final measurements
Input alignment	put texts into a shared representation first	aligned inputs
Anchoring	use bridging observations across languages	model estimates

- do we align the texts before analysis, the outputs after analysis, or the model while it is being estimated?

The right strategy depends on the research question, language expertise, available resources, and what kind of measurement we need.

Separate analysis

Analyze each language-specific subcorpus in its original language. For the housing insecurity example:

- German speeches with German housing terms
- Dutch speeches with Dutch housing terms
- English speeches with English housing terms

Then compare the final speech-level or party-level rates.

```
Multilingual corpus
       |
       v
split by language
       |
       +-- German  -> housing measure
       +-- Dutch   -> housing measure
       +-- English -> housing measure
       |
       v
combine measurements
```

Input alignment

Input alignment first converts multilingual text into a shared representation. The analysis is then run jointly.

Route	How alignment happens
Full-text translation	translate each document into one target language
Token translation	translate the vocabulary after tokenization
Multilingual embeddings	represent documents in a shared vector space

Machine translation as input alignment

Machine translation is the most intuitive form of input alignment:

```
German speech about Miete -> English translation -> English DFM -> model
Dutch speech about huur   -> English translation -> English DFM -> model
English speech about rent ------------------------> English DFM -> model
```

Risk	Validation question
standardized style	do translated texts become artificially similar?
lost ambiguity	are important contextual meanings flattened?
wrong terminology	are policy terms, slogans, or insults mistranslated?

De Vries et al. (2018): Google translation can work for many bag-of-words analyses. Licht et al (2026) replicate this for open source MT systems.

Anchoring

Anchoring uses bridging observations to help place measurements from different languages into a shared space.

Possible bridge	Housing insecurity example
parallel texts	the same housing policy text in multiple languages
bilingual dictionaries	*rent*, *huur*, and *Miete* linked as related terms
comparable documents	debates known to concern housing costs or homelessness

Licht and Lind (2023) discuss anchoring mainly in relation to multilingual topic models: bridging observations constrain the model so that conceptually similar documents receive similar measurements.

Comparing multilingual strategies

Strategy	Advantage	Main risk	Validation focus
Separate analysis	preserves original-language meaning	outputs may not be aligned	do language-specific instruments measure the same concept?
Input alignment	enables one common pipeline	alignment choices become measurement choices	do aligned inputs preserve the meaning needed for analysis?
Anchoring	builds comparability into the model	bridges may be unavailable or weak	do anchors connect conceptually similar texts?

Shared multilingual models do not guarantee comparable inputs

- Multilingual language models can translate texts or embed documents in a shared vector space
- This lowers practical barriers to comparative research
- But models may align some languages and contexts better than others
- High-resource languages, standard varieties, and dominant political contexts are often better represented

Transformation: political speech to text

- Much political communication is spoken:
  - parliaments, rallies, interviews, podcasts, campaign videos
- **Problem**: many sources are not professionally transcribed
- Automatic speech recognition (ASR) turns audio into analyzable text
- This opens new sources, but also creates a new intermediate data product
- It also changes the corpus question: whose speech is recorded, audible, archived, and transcribed?

Validating ASR

Word error rate:

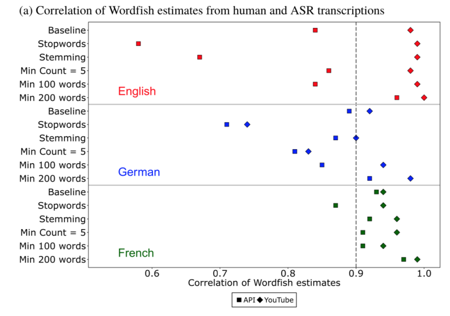
$$WER = \frac{S + D + I}{N}$$

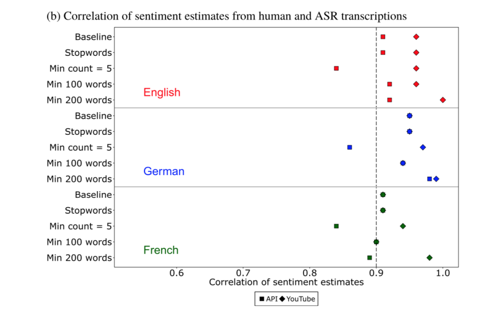
where substitutions, deletions, and insertions are compared with a gold-standard transcript.

- But WER is not enough. Some errors matter more than others.
- Proksch, Wratil, and Wäckerle (2019): test whether ASR errors affect the QTA quantity of interest, not only whether transcripts look clean.

Type	**Gold**	**ASR**	**Why it matters**
Substitution	net zero	next zero	policy vocabulary
Deletion	not support	support	reverses meaning
Insertion	Rishi Sunak	Rishi Sunak said	adds content

ASR and downstream quantities





- Even when transcripts contain errors, aggregate text models can sometimes be robust.

Annotation as representation-building

NLP annotation adds model-generated structure to text before the main analysis.

> `speech text -> token table with lemmas, POS tags, dependencies -> features`

- It usually does not produce the final social science measure
- It helps build the representation that later models or comparisons use

Example: if we build a verb-only DFM, POS-tagging errors become representation errors.

What annotation adds

token	lemma	UPOS	dependency relation
Families	family	NOUN	nsubj
need	need	VERB	root
affordable	affordable	ADJ	amod
homes	home	NOUN	obj
today	today	ADV	advmod

- Lemmas reduce sparsity, but can remove distinctions
- POS tags allow targeted features: nouns, verbs, adjectives
- Dependency relations ask grammatical questions: who acts on what?

Validating annotation

Because annotation sits between the corpus and the analysis, validation should focus on the representation it creates.

Question	Practical check
Are the annotations plausible?	inspect tokens, lemmas, POS tags, and dependencies
Are errors systematic?	check named entities, parliamentary language, rare terms
Do errors affect features?	compare all-word, noun-only, verb-only, or lemma-based DFMs

- validate annotation by asking whether it changes the downstream quantity of interest

LLM coding as interpretation

LLMs can turn unstructured text into structured labels. These labels often require the model to interpret the substantive meaning of the text.

- The model is asked to connect text to a concept.

Input	LLM-generated output
one speech	issue label
one paragraph	frame category
one article segment	economic sentiment label
one manifesto speech	policy topic label

McLaren et al. (2026)

McLaren et al. (2026) evaluate LLM-based political text annotation across different tasks, models, model sizes, learning approaches, and prompt styles.

Common shortcut	What the paper shows
use a larger model	size is an unreliable guide to performance or cost
use few-shot prompting	examples can help, do nothing, or hurt
add persona or chain-of-thought prompting	effects are inconsistent and sometimes negative
pick one good model	the best model varies by task

- LLM labels are a measurement product. Pipeline choices are researcher degrees of freedom.

Codebook and prompt as measurement instruments

Codebook

- concept definition - unit of analysis - response categories - inclusion and exclusion rules - examples

The LLM is not asked to "interpret freely". It is asked to apply a structured coding scheme.

```
Task: classify the policy topic of this speech.

Use only one of the following labels:

economy; welfare; external relations;
freedom and democracy; political system;
fabric of society; social groups; no domain

Return JSON with one key: "label".
```

Other pipeline choices – model, examples, prompt style, output format – can also change the final labels, cost, and reproducibility.

Validation-first LLM labeling

McLaren et al. recommend treating LLM labeling like a measurement pipeline that must be validated before large-scale use.

Step	Practical move
Define the task	freeze the codebook before benchmarking
Create validation data	human-code a stratified sample
Benchmark candidates	compare model families with simple zero-shot prompts
Tune carefully	test few-shot or prompt styles only if needed
Hold out evidence	report final performance on unused test data

- No portable best practice replaces task-specific validation.
- Report the model, version, full prompt or template, sampling settings, zero-shot/few-shot design, examples, hardware or API setup, and efficiency where relevant.

One shared validation logic

Data product	Type	Where to validate	Core question
translation	transformation	input text	is meaning preserved?
transcript	transformation	input text	what errors occur?
POS/lemma/dependency tag	annotation	representation	do tags create useful features?
LLM label or extraction	interpretation	model output / measure	does it match expert judgment?

When results look surprising, ask whether they reflect the political phenomenon or the data-generating pipeline.

Takeaways

- New forms of data make new research questions possible
- But model-generated data are not neutral observations
- Document the pipeline clearly enough that someone else can understand what was transformed, annotated, or interpreted, by which model, and why it was trusted